Proficient in Creo (80 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	**INTRODUCTION TO CREO PARAMETRIC** **Basic Terminologies:** CAD, CAM, CAE, PDM and Design Process **Introduction:** Creo Parametric Basic Modeling Process **Understanding Creo Parametric Concepts:** Understanding Solid Modeling Concepts Understanding Feature-Based Concepts Understanding Parametric Concepts Understanding Associative Concepts Understanding Model-Centric Concepts Recognizing File Extensions **Using the Creo Parametric Interface** Understanding the Main Interface Understanding the Folder Browser Understanding the Web Browser Understanding the Ribbon Interface Setting the Working Directory and Opening and Saving Files Managing Files in Creo Parametric **Creating Sketcher Geometry:** Reviewing Sketcher Theory Understanding Design Intent Modifying the Sketcher Display Utilizing Constraints Sketching with on the fly constraints Sketching Lines Sketching Centerlines Sketching Rectangles Sketching Circles Sketching Arcs Sketching Circular Fillets and Chamfers **Using Sketcher Tools** Understanding Construction Geometry Theory Sketching Points Using Geometry Tools within Sketcher Manipulating Sketches within Sketcher Dimensioning Entities within Sketcher Modifying Dimensions within Sketcher Sketcher Conflicts Placing Sections into Sketcher	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for Practice	Practicals	2.0 Hrs
3	Session 3	**SKETCH TOOLS** **Creating Sketches for Features** Creating Sketches ('Sketch' Feature) Specifying the Sketch Setup Utilizing Sketch References Using Entity from Edge within Sketcher **Advanced Sketching** Using Sketched Curves Sketching Ellipses Sketching Elliptical Fillets Sketching Splines Modifying Splines (Basic & Advanced) Importing and Exporting Spline Points Sketching Conics Sketching Text Analyzing Sketcher Convert Options Analyzing Sketcher Dimension Options	Theory/Demo	2.0 Hrs

		Sketcher Diagnostic Tools		
4	Session 4	Exercises included for Practice	Practicals	2.0 Hrs
5	Session 5	**PART MODELING FEATURES** **Creating Extrudes, Revolves, and Ribs:** Creating Solid Extrude Features Adding Taper to Extrude Feature Common Dashboard Options: Extrude Depth Common Dashboard Options: Feature Direction Common Dashboard Options: Thicken Sketch Creating Solid Revolve Features Common Dashboard Options: Revolve Angle Automatically Adding and Removing Material Creating Rib Features **Using the Creo Parametric Interface – Part 2** Understanding Basic Display Options Analyzing Basic 3-D Orientation Understanding the View Manager Creating and Managing View Orientations Creating Style States using the View Manager Understanding Basic Color and Appearance Options Setting Up New Part Models	Theory/Demo	2.0 Hrs
6	Session 6	Exercises included for Practice	Practicals	2.0 Hrs
7	Session 7	**PART MODELING FEATURES** **Selecting and Editing** Understanding Creo Parametric Basic Controls Using Drag Handles and Dimension Draggers Using Keyboard Shortcuts Understanding the Model Tree Selecting Items using Direct Selection Selecting Items using Query Selection Selecting Multiple Components Using the Search Tool Understanding Selection Filters Renaming Objects Editing Features and Regenerating Activating and Editing Models Deleting and Suppressing Items **Creating Datum Features: Planes and Axes** Creating Datum Features Theory Creating Datum Axes Creating Datum Planes **Utilizing Internal Sketches and Embedded Datums** Creating Internal Sketches Creating Embedded Datum Features **Creating Sweeps and Blends** Creating Sweeps with Open Trajectories Creating Sweeps with Closed Trajectories Analyzing Sweep Feature Attributes Creating a Parallel Blend Protrusion or Cut Experimenting with Parallel Blend Attributes Analyzing Parallel Blend Section Tools	Theory/Demo	2.0 Hrs
8	Session 8	Exercises included for Practice	Practicals	2.0 Hrs
		Creating Holes and Shells Common Dashboard Options: Hole Depth Creating Coaxial Holes Creating Linear Holes Creating Radial and Diameter Holes Exploring Hole Profile Options Creating Shell Features **Advanced hole Creation**		

9	Session 9	Creating Standard Holes Creating Sketched Holes Creating On Point Holes Lightweight Holes Cosmetic Threads **Advanced Shells** Analyzing Shell References and Thickness Options Excluding Surfaces from Shells Extending Shell Surfaces Analyzing Shell Corner Options **Creating Patterns** Direction Patterning in the First Direction Direction Patterning in the Second Direction Axis Patterning in the First Direction Axis Patterning in the Second Direction Creating Reference Patterns of Features Creating Reference Patterns of Components Deleting Patterns or Pattern Members	Theory/Demo	2.0 Hrs
10	Session 10	Exercises included for Practice	Practicals	2.0 Hrs
11	Session 11	**ADVANCED PART MODELING FEATURES** **Advanced Blends** Creating a Rotational Blend Protrusion or Cut Analyzing Rotational Blend Attributes **Advanced Selection** Advanced Chain Selection Advanced Surface Selection **Variable Section Sweeps** Creating Variable Section Sweeps using a Constant Section Creating Variable Section Sweeps Normal to Trajectory Creating Variable Section Sweeps using Constant Normal Direction Creating Variable Section Sweeps Normal to Projection Analyzing Horizontal and Vertical Control in a Variable Section Sweep Creating Variable Section Sweeps Utilizing Multiple Trajectories Creating Variable Section Sweeps with Tangent Trajectories Analyzing Variable Section Sweep Trajectory Options and Rules Using Trajpar with Solid Features Using Trajpar and Datum Graphs with Solid Features **Helical Sweeps** Understanding Helical Sweeps Theory Creating Helical Sweeps for Springs Creating Helical Sweeps for Threads Analyzing Helical Sweep Profile and Pitch Variations Utilizing Variable Sections in Helical Sweeps	Theory/Demo	2.0 Hrs
12	Session 12	Exercises included for Practice	Practicals	2.0 Hrs
13	Session 13	**ADVANCED PART MODELING FEATURES** **Swept Blends** Understanding Swept Blend Theory Creating Swept Blends by Selecting Sections Creating Swept Blends by Sketching Sections Analyzing Swept Blend Section Options Analyzing Swept Blend Section Plane Control Analyzing Horizontal and Vertical Control in a Swept Blend Analyzing Swept Blend Tangency Analyzing Swept Blend Options Analyzing Swept Blend Rules **Advanced Datum Features** Creating Datum Graphs Creating Datum Coordinate Systems Creating Points On or Offset from Entities Creating Points at Intersections	Theory/Demo	2.0 Hrs

		Creating Points using an Offset Coordinate System		
		Creating Sketched Points		
		Creating Curves Through a Point or Vertex		
		Creating a Curve Through a Point Array		
		Creating a Curve From File		
		Creating a Curve from a Cross-Section		
		Creating a Curve From Equation		
		Creating Composite Curves		
		Creating a Curve from Curve Intersections		
		Creating a Curve at Surface Intersection		
		Projecting and Wrapping Curves		
		Trimming Curves		
		Creating Offset Curves		
14	Session 14	Exercises included for Practice	Practicals	2.0 Hrs
15	Session 15	**PART EDITING FEATURES**	Theory/Demo	2.0 Hrs
		Advanced Rounds and Chamfers		
		Analyzing Round Profile		
		Analyzing Round Creation Methods		
		Creating Rounds Through Curve		
		Creating Variable Radius Rounds		
		Auto Round		
		Analyzing Advanced Chamfer Dimensioning Schemes		
		Analyzing Chamfer Creation Methods		
		Analyzing Additional Chamfer Types		
		Creating Corner Chamfers		
		Creating Rounds and Chamfers by Reference		
		Analyzing References and Pieces		
		Using Intent Edges for Rounds and Chamfers		
		Using Chamfer Transitions		
		Using Round Transitions		
		Drafts		
		Creating Draft Features		
		Creating Basic Split Drafts		
		Analyzing Draft Hinges and Pull Direction		
		Drafting Intent Surfaces		
		Creating Drafts with Multiple Angles		
		Using the Extend Intersect Surfaces Draft Option		
		Creating Drafts Split at Sketch		
		Creating Drafts Split at Curve		
		Creating Drafts Split at Surface		
		Creating Drafts with Variable Pull Direction		
16	Session 16	Exercises included for Practice	Practicals	2.0 Hrs
17	Session 17	**PART EDITING FEATURES**	Theory/Demo	2.0 Hrs
		Advanced Patterns		
		Creating Dimension Patterns in One Direction		
		Creating Dimension Patterns in Two Directions		
		Creating Rotational Dimension Patterns		
		Unpatterning Group Patterns		
		Understanding Pattern Regeneration Options		
		Creating Fill Patterns		
		Specifying Fill Pattern Settings		
		Creating Curve Patterns		
		Creating Pattern Tables		
		Creating Point Pattern		
		Group, Copy, and Mirror Tools		
		Creating Local Groups		
		Copying and Pasting Features		
		Moving and Rotating Copied Features		
		Mirroring Selected Features		
		Mirroring All Features		

		Creating Mirrored Parts		
		Advanced Reference Management		
		Editing Feature References		
		Replacing Feature References		
		Replacing Sketcher References		
		Replacing Sketcher Geometry		
18	Session 18	Exercises included for Practice	Practicals	2.0 Hrs
19	Session 19	**ADVANCED TOOLS**	Theory/Demo	2.0 Hrs
		Relations and Parameters		
		Understanding Relation Theory		
		Understanding Relation Types		
		Understanding Basic Relation Operators and Functions		
		Understanding Advanced Relation Operators and Functions		
		Exact Relation		
		Creating Parameters		
		Understanding Advanced Parameter Options		
		Creating Relations		
		Creating Relations for Patterns		
		Creating Section Relations		
		Advanced Layers		
		Understanding Layers		
		Creating and Managing Layers		
		Creating Layer Rules		
		Family Tables		
		Understanding Family Table Theory		
		Creating a Family Table		
		Patternizing Family Table Instances		
		Creating a Multi-Level Family Table		
		Editing Family Table Members		
		Reusing Features		
		Creating UDFs		
		Placing UDFs		
20	Session 20	Exercises included for Practice	Practicals	2.0 Hrs
21	Session 21	**ASSEMBLY MODELING**	Theory/Demo	2.0 Hrs
		Assembling with Constraints		
		Understanding Assembly Theory		
		Creating New Assembly Models		
		Understanding Constraint Theory		
		Understanding Assembly Constraint Status		
		Assembling Components using the Default Constraint		
		Constraining Components using Automatic		
		Constraining Components using Distance		
		Constraining Components using Angle Offset		
		Constraining Components using Parallel		
		Constraining Components using Coincident		
		Constraining Components using Normal		
		Constraining Components using Coplanar		
		Constraining Components using Centred		
		Constraining Components using Tangent		
		Constraining Components using Fix		
		Exploding Assemblies		
		Creating and Managing Explode States		
		Creating Offset Lines between Exploded Components		
22	Session 22	Exercises included for Practice	Practicals	2.0 Hrs
		ASSEMBLY MODELING		
		Creating and Using Component Interfaces		
		Understanding Component Interfaces		
		Using a Placing Component Interface		
		Using a Receiving Component Interface		
		Creating a Component Interface using Save as Interface		

		Auto Placing Components		
		Copying and Pasting Components		
		Repeating Component Placement		
		Creating and Using Flexible Components		
		Adding Flexibility to a Component		
		Placing Flexible Components in an Assembly		
		Adding Flexibility to Already Placed Components		
		Using Flexible Parameters		
23	Session 23	**Restructuring and Mirroring Assemblies**	Theory/Demo	2.0 Hrs
		Restructuring an Assembly		
		Creating Mirrored Assemblies		
		Creating Mirrored Components		
		Creating Mirrored Subassemblies		
		Measuring and Inspecting Models		
		Analyzing Model Units and Mass Properties		
		Measuring Models		
		Creating Planar Part Cross-Sections		
		Measuring Global Interference		
		Advanced Reference Management		
		Editing Feature References		
		Replacing Feature References		
		Replacing Sketcher References		
		Replacing Sketcher Geometry		
24	Session 24	Exercises included for Practice	Practicals	2.0 Hrs
		ADVANCED ASSEMBLY MODELING		
		Capturing and Managing Design Intent		
		Handling Children of Deleted and Suppressed Items		
		Reordering Features		
		Inserting Features		
		Redefining Features and Sketches		
		Capturing Design Intent in Sketches & Features		
25	Session 25	Capturing Design Intent in Parts & Assemblies	Theory/Demo	2.0 Hrs
		Using Assembly Features and Shrinkwrap		
		Understanding Assembly Features		
		Creating an Assembly Cut		
		Creating Assembly Holes		
		Creating a Shrinkwrap Feature		
		Creating a Shrinkwrap Model		
		Summarizing Shrinkwrap Features and Models		
26	Session 26	Exercises included for Practice	Practicals	2.0 Hrs
		Creating and Using Assembly Structure and Skeletons		
		Understanding Skeletons		
		Creating Assembly Structure		
		Creating Skeletons for Space Claims		
		Creating Skeletons for Placement References		
27	Session 27	Copying a Model to a Skeleton	Theory/Demo	2.0 Hrs
		Sharing Skeleton Geometry		
		Creating and Placing Models using Skeleton References		
		Creating a Motion Skeleton		
		Creating Bodies for a Motion Skeleton		
		Assigning Connections for a Motion Skeleton		
		Creating Solid Models from a Motion Skeleton		
28	Session 28	Exercises included for Practice	Practicals	2.0 Hrs
		ADVANCED ASSEMBLY MODELING		
		Creating Cross-Sections, Display Styles, and Combined Views		
		Understanding Assembly Cross-Sections		
		Creating Planar Assembly Cross-Sections		
		Creating Offset Assembly Cross-Sections		
		Creating Zone Assembly Cross-Sections		
		Creating Display Styles		

29	Session 29	Creating Combination Views **Substituting Components By Rep, Envelope, and Model** Understanding Envelopes Creating and Using a Surface Subset Shrinkwrap Envelope Creating and Using a Faceted Shrinkwrap Envelope Creating and Using a All Solid Surfaces Envelope Creating and Using a Create Features Envelope Creating and Using an Envelope Copied from an Existing Part Substituting Components Using User Defined Substituting Components Using By Interchange and Family Table **Understanding Advanced Simplified Rep Functionality** Searching for Components for Simplified Reps Creating Simplified Reps by Size Creating Simplified Reps using Zones Creating Simplified Reps by Distance Creating Simplified Reps using Exterior Components Defining Simplified Reps using Rules Using On-Demand Simplified Reps Creating External Simplified Reps	Theory/Demo	2.0 Hrs
30	Session 30	Exercises included for Practice	Practicals	2.0 Hrs
31	Session 31	**SURFACE MODELING** **Surface Features** Creating Surface features Editing Surface Features in Creo Parametric Copying the Surfaces Mirroring the Surfaces Trimming the Surface Merging the Surfaces Creating the Fill Surface Creating the Intersecting Curves Creating an Offset Surface Giving Thickness to a Surface Converting a Surface into a Solid	Theory/Demo	2.0 Hrs
32	Session 32	Exercises included for Practice	Practicals	2.0 Hrs
33	Session 33	**SHEET METAL DESIGN** **Introduction to sheet metal design** The sheetmetal design environment Sheetmetal parts Design approach **Primary walls** Wall Types Creating the first wall **Secondary and unattached walls** Types of seconday walls Creating unattached walls Secondary walls with relief **Bend features** Bend features Bend lines **Unbend, bend back and cuts** The unbend regular feature The bend back feature Creating sheetmetal cuts **Notches and punches** Creating a punch or notch UDF Placing a punch or notch feature **Sheetmetal forms** Creating form features Creating form parts Placing form features	Theory/Demo	2.0 Hrs

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		Returning the model to the flat state		
34	Session 34	Exercises included for Practice	Practicals	2.0 Hrs
35	Session 35	**SHEET METAL DESIGN** **Unbending sheetmetal geometry** Unbending geometry after defiing the model Deformation areas **Converting solid parts** Creating sheetmetal parts from solid parts Creating a developable part **Sheetmetal drawings with flat states and bend order table** Flat status Creating multimodel drawings Documenting the bend order **Additional features** Flat pattern – Previewing and Creating Solid features Using projected datum curves **Setting up for design** Calculating the developed length Setting up a default radius Setting default fixed geometry **Flexible modeling** Understanding flexible modeling basics Applying selection and tools Utilizing editing and transformations Working with recognition Using propagation and other editing features	Theory/Demo	2.0 Hrs
36	Session 36	Exercises included for Practice	Practicals	2.0 Hrs
37	Session 37	**Creating Drawing Views** Analyzing Drawing Concepts and Theory Analyzing Basic 2-D Orientation Creating New Drawings and Applying Formats Creating and Orienting General Views Adding Drawing Models and Sheets Creating Projection Views **Creating Drawing Views** Creating Cross-Section Views Creating Detailed Views Creating Auxiliary Views Creating 2D Cross Section Views Creating New Drawings using Drawing Templates Modifying Drawing Views Creating Assembly and Exploded Views	Theory/Demo	2.0 Hrs
38	Session 38	Exercises included for Practice	Practicals	2.0 Hrs
39	Session 39	**DIMENSIONS AND ANNOTATIONS** **Creating Drawing Details** Analyzing Detail Concepts and Types Showing and Erasing Detail Items Cleaning Up Dimensions Manipulating Dimensions Creating Driven Dimensions **Creating Drawing Details** Creating Notes Showing a Bill of Materials Analyzing Drawing Associativity	Theory/Demo	2.0 Hrs
40	Session 40	Exercises included for Practice	Practicals	2.0 Hrs